Alexandria Schmid

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Academic Positions

University of Utah, David Eccles School of Business — Salt Lake City, UT
Assistant Professor, Department of Operations & Information Systems — July 2026 – Present

Education

Massachusetts Institute of Technology — Cambridge, MA
PhD in Operations Research — Aug. 2020 – May 2026
Advisor: Alexandre Jacquillat

Georgia Institute of Technology — Atlanta, GA
B.S. Industrial & Systems Engineering — Aug. 2012 – May 2016

Publications

- C. Barnhart, A. Jacquillat, and A. Schmid, "Robotic warehousing operations: a learning-enhanced large-scale neighborhood search approach". *INFORMS Journal on Optimization*.

Working Papers

- B. Martin-Iradi, A. Schmid, K. Cummings, and A. Jacquillat, "A double decomposition algorithm for network planning and operations in deviated fixed-route microtransit". Major revision at *Operations Research*.
- A. Jacquillat, A. Schmid and K. Wang, "Optimizing relay operations toward sustainable logistics". Major revision at *Manufacturing & Service Operations Management*.
- T. Cao, Huang Y., A. Jacquillat, A. Schmid, and L. Zhao. "Service network design in intra-city express delivery".
- C. Barnhart, F. Cordera, A. Jacquillat, A. Schmid, "Order picking in large-scale robotic mobile fulfillment systems".

Awards and Fellowships

- **TSL Dissertation Award** 2026
- Finalist for **TSL Student Paper Prize Competition** 2025
- **Goodwin Medal**, MIT's highest teaching award for a graduate instructor "who performed above and beyond the norm, and whose teaching efforts can truly be characterized as conspicuously effective" 2024
- **Dick and Jerry Smallwood Fellowship**, for graduate students who are applying mathematical models to problems of sustainability and climate 2024 TY6
- **President's Undergraduate Research Award** 2015
- **Stamps President's Scholarship** 2012

Invited Talks

2026 Georgia Tech (Industrial and Systems Engineering)
Auburn University (Industrial and Systems Engineering)
US Naval Academy (Mathematics)

2025 University of Utah (Operations and Information Systems)
University of Pennsylvania, The Wharton School (Operations, Information, and Decisions)
Imperial College London (Analytics and Operations)
Cornell Young Researchers Workshop
ISyE-MS&E-IOE Rising Stars Workshop

Teaching Experience

- **Universal AI** (Online course material through MIT Learn) May 2025 - Aug. 2025
  Teaching Assistant for "AI and Sustainability: Transportation"
  - Designed and taught online recitation modules on integer optimization and transportation applications, including new case studies on microtransit operations and shared bicycle infrastructure

- **Analytics Edge** (15.071) Feb. 2025 - May 2025
  Teaching Assistant
  - Designed and taught weekly recitations, prepared assignments, held office hours, and supervised final projects for a class of MBA students. Student rating: **6.5/7.0**

- **Optimization Methods** (15.093) Sept. 2023 - Dec. 2023
  Teaching Assistant
  - Designed and taught recitation sessions, held office hours, and supervised final projects for a class of undergraduate, masters, and PhD students. Student rating: **6.9/7.0**
  - Introduced poll questions to create opportunities for interaction in large (100+ students) recitation sessions, which have been incorporated into subsequent iterations of the class

- **Computing for Optimization and Statistics** (15.S60)
  Instructor of Record Jan. 2023
  - Coordinated an eight session course on computational research pipeline design: data and visualization in R, machine learning in Python, optimization in Julia, high-performance computing, and version control. Student rating: **4.9/5.0**
  - Designed and taught sessions on Git and high-performance computing

  Session Instructor Jan. 2022
  - Designed and taught a workshop on Git, Github, and distributing computing. Student rating: **6.9/7.0**

- **Integer Programming and Combinatorial Optimization** (15.083) Jan. 2022 - May 2022
  Teaching Assistant
  - Designed and taught weekly recitations, held office hours, and supervised final projects for a class of PhD students. Student rating: **6.9/7.0**

Industry Experience

- **Amazon Robotics** North Reading, MA
  Research Science Intern May 2023 - Aug. 2023
  - Developed a robust optimization model to schedule robots to deliver work to stations, with dependence between tasks and highly variable robot travel times
  - Solved the model via an adversarial scenario generation algorithm, providing efficiency benefits of 10-20% over baseline heuristics in a high-fidelity simulation

- **The Home Depot** Atlanta, GA
  Senior Analyst - Supply Chain Analytics May 2016 - Aug. 2020
  - Wrote order quantity logic for a new in-house replenishment system to unify and replace existing supply chain management systems
  - Designed new order aggregation logic to reduce inventory by $70 million, targeting slow-moving inventory
  - Completed a comprehensive analysis of replenishment system usage and identified company-wide inventory process issues

Service

- Selection committee for the MIT Goodwin Medal 2025 - 2026
- MIT ORC Spring Seminar Co-coordinator 2024
- MIT Teaching Development Fellow 2022 - 2023
- Social and Ethical Responsibilities of Computing Scholar 2022
- **Reviewer:** *Operations Research*, *INFORMS Journal on Computing*, *Transportation Science*, *Production and Operations Management*, *INFORMS Journal on Applied Analytics*